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Electroacoustic transducer, speaker module, and electronic device

Abstract

This application provides an electroacoustic transducer and an electronic device, the electroacoustic transducer includes a central magnet, two side magnets, a lower electrode plate, a side electrode plate, a frame, a sound diaphragm, a voice coil, and two flexible circuit boards. The central magnet and the two side magnets are disposed on the lower electrode plate. The side electrode plate is disposed on a surface of the two side magnets which is opposite to the lower electrode plate. Two surfaces of the frame are respectively fixedly connected to the sound diaphragm and a surface of the side electrode plate which is opposite to the lower electrode plate. The voice coil is located inside the frame. The two flexible circuit boards are symmetrically distributed on a peripheral side of the voice coil, two ends of the flexible circuit board are respectively connected to the voice coil and the side electrode plate.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

(1) This application is a national stage of International Application No. PCT/CN2021/112206, filed on Aug. 12, 2021, which claims priority to Chinese Patent Application No. 202011245180.1, filed on Nov. 10, 2020, and Chinese Patent Application No. 202010866796.4, filed on Aug. 25, 2020. All of the aforementioned applications are hereby incorporated by reference in their entireties.

TECHNICAL FIELD

(2) This application relates to the field of electronic device technologies, and in particular, to an electroacoustic transducer, a speaker module, and an electronic device.

BACKGROUND

(3) A moving coil micro-speaker is an electroacoustic transducer, and is an audio assembly commonly used in a portable electronic device currently. As consumers require increasingly high sound quality of electronic devices, speakers usually need to be designed with larger amplitudes. The speaker includes a voice coil and a magnetic circuit assembly configured to drive the voice coil to vibrate. A driving capability of the magnetic circuit assembly directly affects an amplitude of the speaker, and the driving capability of the magnetic circuit assembly is determined by a size of a magnet disposed in the magnetic circuit assembly. In the conventional technology, a leakage hole is usually designed at a corner location of a speaker. Therefore, magnetic circuit avoidance needs to be performed at the corner location. Consequently, a size of a magnet is small relative to an overall size of the speaker, and a driving capability of a magnetic circuit assembly is further limited, which is not conducive to improving sensitivity of the speaker.

SUMMARY

(4) This application provides an electroacoustic transducer, a speaker module, and an electronic device, to improve sensitivity of the electroacoustic transducer.

(5) According to a first aspect, this application provides an electroacoustic transducer. The electroacoustic transducer includes a central magnet, two side magnets, a lower electrode plate, a side electrode plate, a frame, a sound diaphragm, a voice coil, and two flexible circuit boards. The central magnet is disposed on a surface of the lower electrode plate. The central magnet may be of a rectangular structure, and includes a first corner part and a second corner part. The first corner part and the second corner part are diagonally disposed. The two side magnets and the central magnet are disposed on the same surface of the lower electrode plate, the two side magnets are respectively located on outer peripheral sides of the first corner part and the second corner part, and the two side magnets are disposed centrosymmetrically. A first gap exists between the side magnet and the central magnet. The side electrode plate is disposed on a surface that is of the two side magnets and that is opposite to the lower electrode plate. One surface of the frame is fixedly connected to a surface that is of the side electrode plate and that is opposite to the lower electrode plate, and the other surface is fixedly connected to a periphery of the sound diaphragm, so that the side electrode plate and the sound diaphragm are fastened to each other. The voice coil is located inside the frame, and the voice coil may be of a rectangular ring structure. One end of the voice coil is fixedly connected to the sound diaphragm, and the other end may be inserted into the first gap, so that the voice coil can vibrate under an effect of a magnetic field generated by the central magnet and the side magnet during power-on. The two flexible circuit boards are symmetrically distributed on a peripheral side of the voice coil, one end of the flexible circuit board is fixedly connected to the voice coil, and the other end is fixedly connected to the side electrode plate.

(6) Compared with a conventional solution in which a side magnet is arranged on a side surface of a central magnet, in the electroacoustic transducer in the foregoing solution, the two side magnets are disposed around the two diagonally arranged corner parts of the central magnet, so that a

magnet loss on outer peripheral sides of the two corner parts can be filled, and an overall size of the magnet is increased. Therefore, magnetic induction intensity of the electroacoustic transducer can be effectively improved, and a driving capability of the electroacoustic transducer is improved, so that the electroacoustic transducer has better sensitivity.

(7) During specific disposition, the central magnet and the two side magnets may be separately fastened to a top surface of the lower electrode plate by bonding, to reduce assembly difficulty.

(8) In some possible implementation solutions, the electroacoustic transducer further includes a central electrode plate. The central electrode plate is disposed on a surface that is of the central magnet and that is opposite to the lower electrode plate. In this way, a complete magnetic circuit can be formed between the central magnet and the side magnet, to ensure force-bearing and vibration of the voice coil.

(9) In some possible implementation solutions, the central magnet includes a first side surface, a second side surface, a third side surface, and a fourth side surface. The first side surface and the second side surface are adjacent to the first corner part, and the third side surface and the fourth side surface are adjacent to the second corner part. The side magnet includes a first branch and a second branch. The first branch and the second branch may be connected to form an L-shaped structure. During specific disposition, the first branch and the second branch of one of the two side magnets are respectively located on outer sides of the first side surface and the second side surface, and the first branch and the second branch of the other one of the two side magnets are respectively located on outer sides of the third side surface and the fourth side surface. The two side magnets are set to L-shaped structures, so that not only the magnet loss on the outer peripheral sides of the first corner part and the second corner part can be filled, but also a side magnet is disposed around each side of the central magnet. Therefore, a size of the magnet can be significantly increased, and the electroacoustic transducer can obtain better sensitivity.

(10) In some possible implementation solutions, the central magnet further includes a third corner part and a fourth corner part that are diagonally disposed. The third corner part is connected between the first side surface and the fourth side surface, and the fourth corner part is connected between the second corner part and the third corner part. In this case, the first branch of one of the two side magnets extends between the first corner part and the third corner part, and the second branch of one of the two side magnets extends to an end that is of the second side surface and that is close to the fourth corner part. The first branch of the other one of the two side magnets extends between the second corner part and the fourth corner part, and the second branch of the other one of the two side magnets extends to an end that is of the fourth side surface and that is close to the third corner part, so that the size of the magnet can be further increased.

(11) In some possible implementation solutions, the voice coil includes a third corner corresponding to the third corner part of the central magnet and a fourth corner corresponding to the fourth corner part of the central magnet. The two flexible circuit boards are respectively disposed on outer peripheral sides of the third corner and the fourth corner, one end of the flexible circuit board is fixedly connected to the voice coil, and the other end is fixedly connected to a surface that is of the side electrode plate and that faces the lower electrode plate. In this solution, the two side magnets and the two flexible circuit boards are approximately disposed in a staggered manner. When peripheral space of the voice coil is fully utilized, force uniformity of the sound diaphragm can be further improved, so that a sound effect of the electroacoustic transducer can be improved.

(12) When the flexible circuit board is specifically disposed, the flexible circuit board may include a first connection part, a second connection part, and a third connection part that are sequentially connected. The first connection part is fixedly connected to the voice coil, the second connection part is located on a side that is of the first connection part and that is away from the voice coil, the third connection part is located on a side that is of the second connection part and that is away from the first connection part, and the third connection part is fixedly connected to the side electrode

plate. The side electrode plate is provided with a first notch and a second notch that are symmetrically distributed. The first notch may expose the first connection part and a part of the second connection part of one of the two flexible circuit board, and the second notch may expose the first connection part and a part of the second connection part of the other one of the two flexible circuit board, to provide avoidance space for movement of the flexible circuit board, and avoid interference between the flexible circuit board and the side electrode plate.
(13) In some possible implementation solutions, on an outer peripheral side of the first side surface and an outer peripheral side of the third side surface, an end that is of the second connection part of the corresponding flexible circuit board and that is connected to the third connection part is stacked on the first branch, and an avoidance groove is disposed at a location that is on the first branch and that corresponds to an end part of the second connection part. By using this disposition, the avoidance groove may provide avoidance space for vibration of the second connection part in a direction facing the lower electrode plate, to avoid interference caused by the first branch to movement of the second connection part.
(14) In addition, avoidance holes may also be respectively disposed at locations that are of the side electrode plate and that correspond to the two avoidance grooves. During specific disposition, the two avoidance holes may be connected to the first notch and the second notch respectively. In this way, the avoidance hole may provide avoidance space for vibration of the second connection part in a direction opposite to the lower electrode plate, to avoid interference caused by the side electrode plate to movement of the second connection part.
(15) In some possible implementation solutions, the electroacoustic transducer may further include two auxiliary diaphragms. The two auxiliary diaphragms are respectively disposed on sides that are of the two flexible circuit boards and that face the sound diaphragms. One end of the auxiliary diaphragms may be fixedly connected to the first connection part of the corresponding flexible circuit board, and the other end may be fixedly connected to the third connection part of the corresponding flexible circuit board. In this way, when the sound diaphragm vibrates, the auxiliary diaphragm also vibrates synchronously driven by the flexible circuit board, so that rolling vibration of the voice coil can be suppressed, and the electroacoustic transducer can obtain better sound quality.
(16) In some possible implementation solutions, the sound diaphragm includes a vibration part, a cross-sectional shape of the vibration part of the sound diaphragm is an arc shape, and the vibration part of the sound diaphragm protrudes in a direction opposite to the lower electrode plate. The auxiliary diaphragm also includes a vibration part, a cross-sectional shape of the vibration part of the auxiliary diaphragm is an arc, and the vibration part of the auxiliary diaphragm protrudes in a direction facing the lower electrode plate. In other words, the vibration part of the sound diaphragm and the vibration part of the auxiliary diaphragm protrude in different directions, to release space below the sound diaphragm, and allow the central magnet and the side magnet below the sound diaphragm to be set with a larger height. Therefore, magnetic induction intensity of the electroacoustic transducer is improved, and sensitivity of the electroacoustic transducer is improved.
(17) In some possible implementation solutions, a third notch and a fourth notch may be respectively disposed at locations that are of the lower electrode plate and that correspond to the two auxiliary diaphragms, to provide avoidance space for vibration of the auxiliary diaphragm in a direction facing the lower electrode plate, and avoid interference caused by the lower electrode plate to movement of the auxiliary diaphragm.
(18) In some possible implementation solutions, the electroacoustic transducer may further include a connection frame. The connection frame is located between the voice coil and the sound diaphragm. One end of the connection frame is fixedly connected to the voice coil, and the other end is fixedly connected to the sound diaphragm. The connection frame may be disposed to separate the voice coil from the sound diaphragm, so that the sound diaphragm is away from the

voice coil. In addition, the connection frame may further perform heat dissipation for the voice coil, thereby reducing a risk of damaging the sound diaphragm due to overheating of the voice coil.
(19) According to a second aspect, this application further provides a speaker module. The speaker module includes a box and the electroacoustic transducer in any one of the foregoing possible implementation solutions. The electroacoustic transducer is disposed in the box, and the box is provided with a sound outlet hole connecting an inner side of the box to an outer side of the box, so that a sound emitted by the electroacoustic transducer can be transmitted to the outer side of the box through the sound outlet hole. Because a driving capability of the magnet of the electroacoustic transducer is improved, the speaker module has high sensitivity.
(20) According to a third aspect, this application further provides an electronic device. The electronic device includes a housing and the speaker module in the foregoing implementation solutions. The speaker module is disposed in the housing, a speaker hole is disposed on the housing, and the speaker hole connects the sound outlet hole to an outer side of the electronic device, so that a sound emitted by an electroacoustic transducer can be transmitted to the outer side of the electronic device through the sound outlet hole and the speaker hole. Because the speaker module has high sensitivity, sound quality of the electronic device is also improved.
(21) According to a fourth aspect, this application further provides an electronic device. The electronic device includes a housing, a display module, and a receiver. The receiver is disposed in the housing, and the receiver may be the electroacoustic transducer in the foregoing implementation solutions. The display module includes a glass cover and a display panel. The glass cover is fastened to a side of the housing, and the display panel is fastened to an inner surface that is of the glass cover and that faces the housing. The electronic device is provided with a receiver hole, so that a sound emitted by the receiver can be transmitted to an outer side of the electronic device through the receiver hole. During specific disposition, the receiver hole may be disposed on the cover, or the receiver hole may be formed between an edge of the cover and the housing, or the receiver hole may be disposed on the housing. Because the speaker module has high sensitivity, sound quality of the electronic device is also improved.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) FIG. **1** is a schematic diagram of a structure of an electronic device **1** according to an embodiment of this application;
(2) FIG. **2** is a schematic diagram of a structure of a speaker module of the electronic device shown in FIG. **1**;
(3) FIG. **3** is a schematic exploded view of the speaker module shown in FIG. **2**;
(4) FIG. **4** is a schematic diagram of a structure of an electroacoustic transducer according to an embodiment of this application;
(5) FIG. **5** is a partial schematic exploded view of the electroacoustic transducer shown in FIG. **4**;
(6) FIG. **6** is a partial schematic exploded view of a vibration assembly of the electroacoustic transducer shown in FIG. **4**;
(7) FIG. **7** is a sectional view of a vibration assembly of the electroacoustic transducer shown in FIG. **4** in an A-A direction;
(8) FIG. **8** is a partial schematic exploded view of a magnetic circuit assembly of the electroacoustic transducer shown in FIG. **4**;
(9) FIG. **9** is a schematic diagram of a partial structure of the electroacoustic transducer shown in FIG. **4**;
(10) FIG. **10** is a sectional view of the electroacoustic transducer shown in FIG. **4** in an A-A direction; and

(11) FIG. **11** is a sectional view of the electroacoustic transducer shown in FIG. **4** in a B-B direction.

REFERENCE NUMERALS

(12) **1**—Electronic device; **100**—Housing; **200**—Display module; **300**—Receiver; **400**—Camera module; **500**—Speaker module; **600**—Mainboard; **700**—Battery; **110**—Middle frame; **120**—Rear cover; **1001**—Speaker hole; **210**—Glass cover; **220**—Display panel; **211**—Light transmission region; **212**—Receiver hole; **510**—Electroacoustic transducer; **520**—Box; **530**—Cover body; **540**—Circuit board; **521**—Sound outlet hole; **522**—Slot; **523**—Protrusion; **10**—Frame; **20**—Vibration assembly; **30**—Magnetic circuit assembly; **21**—Sound diaphragm; **22**—Connection frame; **23**—Voice coil; **24**, **24***a*, **24***b*—Flexible circuit board; **25**—Auxiliary diaphragm; **211**—Diaphragm; **212**—Dome; **213**—First fastening part of the diaphragm; **214**—Vibration part of the diaphragm; **215**—Second fastening part of the diaphragm; **221**—Body; **222**—Extension part; **241**—First connection part; **242**—Second connection part; **243**—Third connection part; **2411**—Arc segment of the first connection part; **2412**—Straight line segment of the first connection part; **2421**—Arc segment of the second connection part; **2422**—Straight line segment of the second connection part; **251**—First fastening part of the auxiliary diaphragm; **252**—Vibration part of the auxiliary diaphragm; **253**—Second fastening part of the auxiliary diaphragm; **31**—Central electrode plate; **32**—Side electrode plate; **33**—Central magnet; **34**, **34***a*, **34***b*—Side magnet; **35**—Lower electrode plate; **36**—First gap; **341**—First branch; **342**—Second branch; **331**—First side surface; **332**—Second side surface; **333**—Third side surface; **334**—Fourth side surface; **37**—Connection space; **38**—Second gap; **321**—First notch; **322**—Second notch; **351**—Third notch; **352**—Fourth notch; **343**—First avoidance groove; **323**—Second avoidance groove; **344**—Third avoidance groove; **324**—Fourth avoidance groove.

DETAILED DESCRIPTION OF ILLUSTRATIVE EMBODIMENTS

(13) The following clearly and completely describes the technical solutions in the implementations of this application with reference to the accompanying drawings in the implementations of this application.

(14) An embodiment of this application provides an electroacoustic transducer. The electroacoustic transducer is configured to convert an electrical signal into a sound signal. In the electroacoustic transducer, structures and relative locations of a magnetic circuit assembly and a flexible circuit board are optimized. Therefore, an area and a volume of a magnet are increased, and a driving capability of the magnetic circuit assembly can be improved, so that the electroacoustic transducer has better sensitivity. An embodiment of this application further provides an electronic device including the electroacoustic transducer. The electronic device may be a product with a sound play function, for example, a mobile phone, a tablet computer, a notebook computer, a wearable device, or a personal stereo. The wearable device may be a smart band, a smart watch, a smart head-mounted display device, smart glasses, or the like. For example, the electroacoustic transducer may be used for the electronic device as a speaker core of a speaker module (also referred to as a loudspeaker), or may be used for the electronic device as a receiver (also referred to as an earpiece).

(15) FIG. **1** is a schematic diagram of a structure of an electronic device **1** according to an embodiment of this application. The electronic device **1** shown in FIG. **1** is described by using a mobile phone as an example. The electronic device includes a housing **100**, a display module **200**, a receiver **300**, a camera module **400**, a speaker module **500**, a mainboard **600**, and a battery **700**. It should be noted that FIG. **1** and the following related accompanying drawings merely schematically show some components included in the electronic device **1**. Actual shapes, actual sizes, actual locations, and actual structures of these components are not limited by FIG. **1** and the following accompanying drawings.

(16) The housing **100** includes a middle frame **110** and a rear cover **120**. The rear cover **120** is fastened to a side of the middle frame **110**. In an implementation, the rear cover **120** may be

fastened to the middle frame **110** by assembling. In another implementation, the rear cover **120** and the middle frame **110** may alternatively be of an integrated structure. A speaker hole **1001** is disposed on the housing **100**, and there may be one or more speaker holes **1001**. For example, there are a plurality of speaker holes **1001**, and the plurality of speaker holes **1001** are disposed on the middle frame **110**. The speaker hole **1001** may connect an inner side of the electronic device **1** to an outer side of the electronic device **1**.

(17) The display module **200** is fastened to a side that is of the middle frame **110** and that is away from the rear cover. In this case, the display module **200** and the rear cover **120** are disposed opposite to each other, and the display module **200**, the middle frame **110**, and the rear cover **120** jointly form the inner side of the electronic device **1**. The display module **200** includes a glass cover **210** and a display panel **220**. The glass cover **210** is stacked on a side that is of the display panel **220** and that is away from the middle frame **110**. The display panel **220** is configured to display an image, and the display panel **220** may further be integrated with a touch function. The glass cover **210** may be disposed against the display panel **220**, and is mainly configured to protect the display panel **220** and prevent against dust. A light transmission region **211** and a receiver hole **212** are disposed on the glass cover **210**. The light transmission region **211** may allow light to pass through. For example, an ink layer of the glass cover **210** is hollowed out in a region in which the light transmission region **211** is located. The receiver hole **212** is a through hole penetrating the glass cover **210**.

(18) In some other embodiments, a receiver hole is formed between an edge of the glass cover **210** and the housing **100**. For example, the receiver hole is formed between an edge, at the top of the electronic device **1**, of the glass cover **210** and an edge, at the top of the electronic device **1**, of the middle frame **110** of the housing **100**. In some other embodiments, a receiver hole is disposed on the housing **100**. For example, a receiver hole is formed in a region, at the top of the electronic device **1**, of the middle frame **110** of the housing **100**. A specific formation structure and location of the receiver hole are not strictly limited in this application.

(19) The receiver **300** is disposed in the housing **100**. The receiver **300** is located between the display module **200** and the rear cover **120**. A sound emitted by the receiver **300** is transmitted to the outer side of the electronic device **1** through the receiver hole **212**, to implement a sound play function of the electronic device **1**. For example, the receiver **300** may be an electroacoustic transducer described in the following embodiments. In another embodiment, the receiver **300** may be alternatively an electroacoustic transducer with another structure.

(20) The camera module **400** is accommodated in the housing **100**. The camera module **400** is located between the display module **200** and the rear cover **120**. The camera module **400** collects light through the light transmission region of the glass cover **210**, to perform photographing. The electronic device **1** may further include another camera module accommodated in the housing **100**. A photographing through hole may be disposed on the rear cover **120**. The another camera module may collect light through the photographing through hole, to perform photographing.

(21) The speaker module **500** is accommodated in the housing **100**. The speaker module **500** is located between the display module **200** and the rear cover **120**. A sound emitted by the speaker module **500** can be transmitted to the outer side of the electronic device **1** through the speaker hole **1001**, to implement the sound play function of the electronic device **1**. The speaker module **500** includes a speaker core. The speaker core may be the electroacoustic transducer **510** described in the following embodiments. In another embodiment, the speaker core may be alternatively an electroacoustic transducer with another structure.

(22) Both the mainboard **600** and the battery **700** are accommodated in the housing **100**. The mainboard **600** is located on one side of the battery **700**. For example, the mainboard **600** is located on an upper part of the electronic device **1**, and the battery **700** is located on a lower part of the electronic device **1**. The battery **700** is configured to supply power to the electronic device **1**. The mainboard **600** may be a rigid mainboard, or may be a flexible mainboard, or may be a

combination of a rigid mainboard and a flexible mainboard. In addition, the mainboard **600** may be configured to set a chip. The chip may be a central processing unit (central processing unit, CPU for short), a graphics processing unit (graphics processing unit, GPU for short), a universal flash storage (universal flash storage, UFS for short), or the like. Functional modules of the electronic device **1**, for example, the display module **200**, the camera module **400**, the speaker module **500**, and the receiver **300**, are coupled to the CPU.

(23) Refer to FIG. **2** and FIG. **3**. FIG. **2** is a schematic diagram of a structure of a speaker module of the electronic device shown in FIG. **1**. FIG. **3** is a schematic exploded view of the speaker module shown in FIG. **2**.

(24) The speaker module **500** includes an electroacoustic transducer **510**, a box **520**, a cover body **530**, and a circuit board **540**. The cover body **530** is disposed on a side of the box **520**, and the cover body **530** are fixedly connected to the box **520** to form a sound box. The electroacoustic transducer **510** is located inside the sound box. One end of the circuit board **540** is located inside the sound box, to connect to the electroacoustic transducer **510**. The other end of the circuit board **540** is located outside the sound box, to electrically connect the electroacoustic transducer **510** to an external component of the speaker module **500**. For example, the end that is of the circuit board **540** and that is located outside the sound box may be fixed and electrically connected to the mainboard.

(25) A sound outlet hole **521** is disposed on a side surface of the box **520**. The sound outlet hole **521** connects an inner side of the sound box to an outer side of the sound box. A sound emitted by the electroacoustic transducer **510** can be transmitted to the outer side of the sound box through the sound outlet hole **521**. With reference to FIG. **1**, the speaker hole **1001** of the housing **100** connects the sound outlet hole **521** of the speaker module **500** to the outer side of the electronic device **1**. The sound emitted by the electroacoustic transducer **510** can be transmitted to the outer side of the electronic device **1** through the sound outlet hole **521** and the speaker hole **1001**.

(26) A slot **522** connecting the inner side of the sound box to the outer side of the sound box is further disposed at an end that is of the box **520** and that faces the cover body **530**. The circuit board **540** may specifically extend to the outer side of the sound box through the slot **522**. The circuit board **540** may be a flexible circuit board. During specific implementation, a protrusion **523** disposed in a direction facing the cover body **530** is disposed in the box **520**. One end of the protrusion **523** is disposed close to the slot **522**, and the other end is disposed close to the electroacoustic transducer **510**. A part that is of the circuit board **540** and that is located in the sound box is disposed on an end face of the protrusion **523**. In this way, the circuit board **540** can be firmly fixed in the sound box, to reduce a risk of damaging the circuit board **540** due to shaking. In addition, an end that is of the circuit board **540** and that is located in the sound box may include two branches. Ends of the two branches are respectively formed as two connection ends connected to the electroacoustic transducer **510**.

(27) Refer to FIG. **4** and FIG. **5**. FIG. **4** is a schematic diagram of a structure of an electroacoustic transducer according to an embodiment of this application. FIG. **5** is a partial schematic exploded view of the electroacoustic transducer shown in FIG. **4**. In this embodiment of this application, an x direction is defined as a length direction of the electroacoustic transducer **510**, a y direction is defined as a width direction of the electroacoustic transducer **510**, and a Z direction is defined as a thickness direction of the electroacoustic transducer **510**. It should be noted that orientation terms such as “top” and “bottom” used for the electroacoustic transducer **510** in this embodiment of this application are mainly intended for description based on a display orientation of the electroacoustic transducer **510** in FIG. **4**, and do not limit an orientation of the electroacoustic transducer **510** in an actual application scenario.

(28) The electroacoustic transducer **510** includes a frame **10**, a vibration assembly **20**, and a magnetic circuit assembly **30**. The vibration assembly **20** and the magnetic circuit assembly **30** are mounted on the frame **10**. A part that is of each component of the vibration assembly **20** and that is

connected to the frame **10** is fixed relative to the frame **10**, and a remaining part may vibrate relative to the frame **10**. The magnetic circuit assembly **30** is fixed relative to the frame **10**, and the magnetic circuit assembly **30** may be configured to provide a driving magnetic field for the vibration assembly **20**.

(29) FIG. **6** is a partial schematic exploded view of a vibration assembly of the electroacoustic transducer shown in FIG. **4**. The vibration assembly **20** of the electroacoustic transducer includes a sound diaphragm **21**, a connection frame **22**, a voice coil **23**, two flexible circuit boards **24**, and two auxiliary diaphragms **25**. The voice coil **23** may be roughly of a rectangular ring structure. For example, the voice coil **23** may be in a rounded rectangular ring shape. The voice coil **23** includes four straight edges and four corner parts, and one corner part is connected between two adjacent straight edges. In this embodiment, the voice coil **23** is inserted into the magnetic circuit assembly. Under an effect of a magnetic field provided by the magnetic circuit assembly, when the voice coil **23** is powered on, an ampere force may be generated, to drive another component of the vibration assembly to vibrate.

(30) The sound diaphragm **21** includes a diaphragm **211** and a dome **212**. The diaphragm **211** is roughly in a rectangular ring shape, and includes a first fastening part **213**, a vibration part **214**, and a second fastening part **215**. The first fastening part **213** is located inside the vibration part **214**, and the second fastening part **215** is located outside the vibration part **214**. The first fastening part **213** of the diaphragm **211** is fixedly connected to the dome **212**, and the second fastening part **215** is fixedly connected to the frame. In addition, during specific disposition, the dome **212** is connected to a top surface of the first fastening part **213**, and the frame is connected to a bottom surface of the second fastening part **215**. A cross-sectional shape of the vibration part **214** of the diaphragm **211** is an arc or approximately arc shape, an extension track of the vibration part **214** is in a rounded rectangle shape, and the vibration part **214** is convex. To be specific, the vibration part **214** protrudes in a direction away from the voice coil **23**. When the vibration part **214** is subject to an external force, the vibration part **214** can deform, so that the first fastening part **213** and the dome **212** move relative to the second fastening part **215**.

(31) The connection frame **22** may be configured to connect the voice coil **23** to the sound diaphragm **21**. Specifically, the connection frame **22** includes a body **221** and an extension part **222**. A shape of the body **221** is roughly a rectangular ring shape matching the voice coil **23**. One end of the body **221** is connected to an end face on a side that is of the voice coil **23** and that faces the sound diaphragm **21**, and the other end of the body **221** is connected to an end face of a side that is of the sound diaphragm **21** and that faces the voice coil **23**. The extension part **222** is disposed at an end that is of the body **221** and that faces the voice coil **23**. When the body **221** is fastened to the voice coil **23**, the extension part **222** is attached to an outer peripheral surface of the voice coil **23**. There may be one or more extension parts **222**. For example, there are two extension parts **222** shown in FIG. **6**, and the two extension parts **222** may be symmetrically disposed near two diagonals of the body **221**. In this way, after the body **221** and the voice coil **23** are assembled, the two extension parts **222** may be respectively attached and fastened to outer peripheral surfaces of the two diagonally disposed corner parts of the voice coil **23**. Therefore, when force uniformity of the voice coil **23** is ensured, connection strength between the connection frame **22** and the voice coil **23** can be improved. The connection frame **22** may be disposed to separate the voice coil **23** from the sound diaphragm **21**, so that the sound diaphragm **21** is away from the voice coil **23**. In addition, the connection frame **22** may further perform heat dissipation for the voice coil **23**, thereby reducing a risk of damaging the sound diaphragm **21** due to overheating of the voice coil **23**.

(32) Refer to FIG. **6** again. Structures of the two flexible circuit boards **24** are the same, and the two flexible circuit boards **24** are centrosymmetrically disposed on the outer peripheral side of the voice coil **23**. For example, the two flexible circuit boards **24** are separately disposed near the two diagonally disposed corner parts of the voice coil **23**. Two ends of a lead of the voice coil **23** may

be electrically connected to the two flexible circuit boards **24** respectively, and the two flexible circuit boards **24** are electrically connected to the foregoing circuit board, to implement electrical connection to an external component of the electroacoustic transducer. Alternatively, two ends of a lead of the voice coil **23** may be electrically connected to one of the two flexible circuit boards **24**, and the flexible circuit board **24** is electrically connected to the foregoing circuit board, to implement electrical connection to an external component of the electroacoustic transducer.

(33) During specific implementation, the flexible circuit board **24** includes a first connection part **241**, a second connection part **242**, and a third connection part **243**. The second connection part **242** is connected between the first connection part **241** and the third connection part **243**. The first connection part **241**, the second connection part **242**, and the third connection part **243** are roughly of strip-shaped structures. The first connection part **241** is connected to an outer peripheral surface of a corresponding corner part of the voice coil **23**. In addition, to increase a contact area between the first connection part **241** and the voice coil **23**, a head end of the first connection part **241** may extend to an outer side surface that is adjacent to the corresponding corner part on the voice coil **23** and that extends in a first direction (that is, the width direction y of the electroacoustic transducer). A head end of the second connection part **242** is connected to a tail end of the first connection part **241**, the second connection part **242** is located on a side that is of the first connection part **241** and that is away from the voice coil **23**, and the second connection part **242** and the first connection part **241** are disposed at an interval. A head end of the third connection part **243** is connected to a tail end of the second connection part **242**, the third connection part **243** is located on a side that is of the second connection part **242** and that is away from the first connection part **241**, and the third connection part **243** and the second connection part **242** are disposed at an interval. For example, a connection location between the first connection part **241** and the second connection part **242** is disposed roughly in a U shape, and a connection location between the second connection part **242** and the third connection part **243** is also disposed roughly in a U shape.

(34) In addition, when the voice coil **23** is of a rounded rectangular structure, to better match a shape of the outer peripheral surface of the voice coil **23**, the first connection part **241** and the second connection part **242** may separately include an arc segment and a straight line segment. A straight line segment **2412** of the first connection part **241** is disposed close to the head end of the first connection part **241**, and an arc segment **2411** of the first connection part **241** is disposed close to the tail end of the first connection part **241**. An arc segment **2421** of the second connection part **242** is disposed close to the head end of the second connection part **242**, and a straight line segment **2422** of the second connection part **242** is disposed close to the tail end of the second connection part **242**. The third connection part **243** is disposed roughly in an L shape, a corner location of the third connection part **243** is located on an outer side of the are segment of the second connection part **242**, and a corner of the third connection part **243** is a rounded corner. In this embodiment, the are segment **2411** of the first connection part **241**, the arc segment **2421** of the second connection part **242**, and the corner of the third connection part **243** are all disposed coaxially with a corresponding corner part of the voice coil **23**. It may be understood that, that two structures are coaxially disposed means that central axes (or referred to as center lines) of the two structures overlap, and a slight deviation caused by a manufacturing tolerance, an assembly tolerance, or the like is allowed.

(35) In this embodiment, because the are segment **2411** of the first connection part **241**, the are segment **2421** of the second connection part **242**, and the corner of the third connection part **243** are coaxially disposed with the corresponding corner part of the voice coil **23**, when the voice coil **23** drives the flexible circuit board **24** to vibrate, a shape of the flexible circuit board **24** can better adapt to deformation and displacement requirements, so that the flexible circuit board **24** has better reliability and a longer service life. In addition, by using this disposition, outside space of the corner part of the voice coil **23** can be fully used at each connection part, to route a longer wire, so that when the flexible circuit board **24** vibrates with the voice coil **23** at a large amplitude, stress is

small, and the flexible circuit board **24** has higher reliability.

(36) FIG. **7** is a sectional view of a vibration assembly of the electroacoustic transducer shown in FIG. **4** in an A-A direction. Refer to FIG. **6** and FIG. **7**. Structures of the two auxiliary diaphragms **25** are the same, and the two auxiliary diaphragms **25** are respectively disposed corresponding to the two flexible circuit boards **24**. Specifically, the two auxiliary diaphragms **25** may be separately disposed on a side of a bottom surface of the corresponding flexible circuit board **24**. The auxiliary diaphragm **25** includes a first fastening part **251**, a vibration part **252**, and a second fastening part **253**. The first fastening part **251** is located inside the vibration part **252**, and the second fastening part **253** is located outside the vibration part **252**. An extension track of the auxiliary diaphragm **25** matches the track of the first connection part **241**. The first fastening part **251** of the auxiliary diaphragm **25** is fixedly connected to the first connection part **241** of the corresponding flexible circuit board **24**. The second fastening part **253** is fixedly connected to the third connection part **243** of the corresponding flexible circuit board **24**. A cross-sectional shape of the vibration part of the auxiliary diaphragm **25** is an arc or approximately arc shape, and the vibration part **252** is concave. To be specific, the vibration part **252** protrudes in a direction away from the flexible circuit board **24**. When the vibration part **252** of the auxiliary diaphragm **25** subject to an external force, the vibration part **252** can deform, so that the first fastening part **251** and the second fastening part **253** of the auxiliary diaphragm **25** move relative to each other.

(37) FIG. **8** is a partial schematic exploded view of a magnetic circuit assembly of the electroacoustic transducer shown in FIG. **4**. The magnetic circuit assembly **30** of the electroacoustic transducer includes a central electrode plate **31**, a side electrode plate **32**, a central magnet **33**, two side magnets **34**, and a lower electrode plate **35**. The central electrode plate **31**, the side electrode plate **32**, and the lower electrode plate **35** are magnetic conductors, and the central magnet **33** and the two side magnets **34** are permanent magnets.

(38) During specific implementation, the central magnet **33** and the two side magnets **34** are separately disposed on a top surface of the lower electrode plate **35**, and each magnet may be fastened to the top surface of the lower electrode plate **35** by bonding. The central magnet **33** is roughly of a rectangular structure. For example, the central magnet **33** may be of a rounded rectangular structure. The central magnet **33** includes four side surfaces and four corner parts, and one corner part is connected between two adjacent side surfaces. The two side magnets **34** are centrosymmetrically disposed on an outer peripheral side of the central magnet **33**, and first gaps **36** are formed between the two side magnets **34** and the central magnet **33**. For example, the two side magnets **34** may be respectively disposed around two corner parts that are of the central magnet **33** and that are diagonally disposed.

(39) The side magnet **34** includes a first branch **341** disposed in a first direction (that is, the width direction y of the electroacoustic transducer) and a second branch **342** disposed in a second direction (that is, the length direction x of the electroacoustic transducer). In other words, the side magnet **34** is disposed roughly in an L shape. During specific implementation, for the side magnet **34***a* located on a left side of the central magnet **33**, the first branch **341** of the side magnet **34***a* is located on an outer side of the first side surface **331** of the central magnet **33**, and the second branch **342** of the side magnet **34***a* is located on an outer side of the second side surface **332** of the central magnet **33**. In other words, the side magnet **34***a* is disposed around a first corner part A between the first side surface **331** and the second side surface **332**. For the side magnet **34***b* located on a right side of the central magnet **33**, the first branch **341** of the side magnet **34***b* is located on an outer side of the third side surface **333** of the central magnet **33**, and the second branch **342** of the side magnet **34***b* is located on an outer side of the fourth side surface **334** of the central magnet **33**. In other words, the side magnet **34***b* is disposed around a second corner part B between the third side surface **333** and the fourth side surface **334**. When the central magnet **33** is of a rounded rectangular structure, a corner at a connection location between the first branch **341** and the second branch **342** is also a rounded corner, and the rounded corner is coaxially disposed with a

corresponding corner part of the central magnet **33**. In this way, a shape of an outer peripheral surface of the side magnet **34** can better match that of the central magnet **33**, so that a magnetic field with uniform strength can be formed in the first gap **36** between the side magnet **34** and the central magnet **33**. It may be understood that the first side surface **331** and the third side surface **333** of the central magnet **33** are respectively side surfaces disposed in a y-axis direction, and the second side surface **332** and the fourth side surface **334** of the central magnet **33** are respectively side surfaces disposed in an x-axis direction.

(40) In addition, on the first side surface **331** of the central magnet **33**, the corresponding first branch **341** extends to a location between the first corner part A and a third corner part C, for example, a location close to a middle region of the first side surface **331**. On the fourth side surface **334** of the central magnet **33**, the corresponding second branch **342** approximately extends to an end that is of the fourth side surface **334** and that is close to the third corner part C. The third corner part C of the central magnet **33** is a corner part between the first side surface **331** and the fourth side surface **334**. Similarly, on the third side surface **333** of the central magnet **33**, the corresponding first branch **341** extends to a location between the second corner part B and a fourth corner part D, for example, a location close to a middle region of the third side surface **333**. On the second side surface **332** of the central magnet **33**, the corresponding second branch **342** approximately extends to an end close to the fourth corner part D. The fourth corner part D of the central magnet **33** is a corner part between the second side surface **332** and the third side surface **333**. In this way, connection spaces **37** are respectively formed on outer sides of the third corner part C and the fourth corner part D of the central magnet **33**, and each connection space **37** separately connects the first gaps **36** between the two magnets **34** on left and right sides and the central magnet **33**, and is connected to an outer side of the magnetic circuit assembly **30**.

(41) Still refer to FIG. **8**. The central electrode plate **31** is located on a side that is of the central magnet **33** and that is away from the lower electrode plate **35**, and the central electrode plate **31** may be specifically fastened to the central magnet **33** by bonding. The side electrode plate **32** is located on a side that is of the two side magnets **34** and that is away from the lower electrode plate **35**. Similarly, the side electrode plate **32** may be fastened to the side magnets **34** by bonding. During specific implementation, the side electrode plate **32** is roughly of a frame structure, and the side electrode plate **32** is disposed around the central electrode plate **31**. A second gap **38** exists between the side electrode plate **32** and the central electrode plate **31**. The second gap **38** is connected to the first gap **36**. On an inner side that is of the side electrode plate **32** and that faces the central electrode plate **31**, a first notch **321** and a second notch **322** are respectively disposed at locations that are of the side electrode plate **32** and that correspond to the two connection spaces **37**, to connect each connection space **37** to an outer side of the magnetic circuit assembly **30** in a positive direction of a z-axis. This increases a size of the connection space **37** in the z-axis direction.

(42) In addition, notches may be disposed at locations that are of the lower electrode plate **35** and that correspond to the two connection spaces **37**, which are respectively a third notch **351** and a fourth notch **352**. The third notch **352** and the fourth notch **352** may connect each connection space **37** to an outer side of the magnetic circuit assembly **30** in a negative direction of the z-axis, to increase the size of the connection space **37** in the z-axis direction.

(43) Refer to FIG. **8** to FIG. **10**. FIG. **9** is a schematic diagram of a partial structure of the electroacoustic transducer shown in FIG. **4**. FIG. **10** is a sectional view of the electroacoustic transducer shown in FIG. **4** in an A-A direction. After the vibration assembly **20**, the frame **10**, and the magnetic circuit assembly **30** are assembled, a peripheral of the sound diaphragm **21** is fastened to a first surface of the frame **10**. For example, the second fastening part **211** of the sound diaphragm **21** may be fastened to the frame **10** by bonding. A vibration direction of the sound diaphragm **21** is parallel to the thickness direction z of the electroacoustic transducer **510**. The voice coil **23** is located inside the frame **10**, and an end of the voice coil **23** is fixedly connected to

the sound diaphragm **21**. For example, the voice coil **23** is fixedly connected to the first fastening part **213** of the sound diaphragm **21** through the connection frame **22**.

(44) The central magnet **33** and the two side magnets **34** are fastened to a side that is of the lower electrode piece **35** and that faces the frame body **10**. One side magnet **34**_a_ is disposed around the first corner part A of the central magnet **33**, and the other side magnet **34**_b_ is disposed around the second corner part B of the central magnet **33**. An end that is of the voice coil **23** and that is away from the sound diaphragm **21** is partially inserted into the first gap **36** formed by the central magnet **33** and the side magnet **34**.

(45) An edge of a surface that is of the side electrode plate **32** and that faces the sound diaphragm **21** is fixedly connected to a second surface of the frame **10**, and a surface that is of the side electrode plate **32** and that is opposite to the sound diaphragm **21** is fixedly connected to the two side magnets **34**. The central electrode plate **31** is located inside the side electrode plate **32**, and the central electrode plate **31** is fastened to a surface that is of the central magnet **33** and that faces the sound diaphragm **21**. The end that is of the voice coil **23** and that is away from the sound diaphragm **21** is sequentially inserted into the second gap **38** and the first gap **36** in the negative direction of the z-axis.

(46) The two flexible circuit boards **24** are located on a side that is of the side electrode plate **32** and that is away from the frame **10**, and the two flexible circuit boards **24** are respectively disposed near the two diagonally disposed corner parts of the voice coil **23**. For example, the first connection part **241** of the flexible circuit board **24**_a_ is fixedly connected to an end that is of an outer peripheral surface of a third corner C**1** of the voice coil **23** and that is away from the sound diaphragm **21**, and the third connection part **243** of the flexible circuit board **24**_a_ is fixedly connected to an edge of a surface that is of the side electrode plate **32** and that is away from the frame **10**. The first connection part **241** of the flexible circuit board **24**_b_ is fixedly connected to an end that is of an outer peripheral surface of a fourth corner D**1** of the voice coil **23** and that is away from the sound diaphragm **21**, and the third connection part **243** of the flexible circuit board **24**_b_ is fixedly connected to an edge of a surface that is of the side electrode plate **32** and that is away from the frame **10**. On a surface that is of the side electrode plate **32** and that faces the frame **10**, the first notch **321** of the side electrode plate **32** may expose the first connection part **241** and the second connection part **242** of the flexible circuit board **24**_a_, and the second notch **322** of the side electrode plate **32** may expose the first connection part **241** and the second connection part **242** of the flexible circuit board **24**_b_. In other words, on an xy plane, the two side magnets **34** and the two flexible circuit boards **24** are approximately disposed in a staggered manner, the two side magnets **34** occupy outer peripheral regions of one pair of corners (that is, a first corner A**1** and a second corner B**1** of the voice coil) of the voice coil **23**, and the two flexible circuit boards **24** occupy outer peripheral regions of the other pair of corners (that is, the third corner C**1** and the fourth corner D**1** of the voice coil) of the voice coil **23**.

(47) It should be noted that, in the foregoing solution, the voice coil **23** is disposed around an outer peripheral side of the central magnet **33**, and the first corner A**1**, the second corner B**1**, the third corner C**1**, and the fourth corner D**1** of the voice coil **23** respectively correspond to the first corner part A, the second corner part B, the third corner part C, and the fourth corner part D of the central magnet **33**.

(48) The two auxiliary diaphragms **25** are separately fastened on a surface that is of the two flexible circuit boards **24** and that is away from the side electrode plate **32**. The first fastening part **251** of the auxiliary diaphragm **25** is fixedly connected to the first connection part **241** of the corresponding flexible circuit board **24**. The second fastening part **253** is fixedly connected to the third connection part **243** of the corresponding flexible circuit board **24**. The vibration part **252** of the auxiliary diaphragm **25** and the second connection part **242** of the corresponding flexible circuit board **24** are at opposite positions. In this case, the two auxiliary diaphragms **25** are respectively located in the two connection spaces **37** of the magnetic circuit assembly **30**.

(49) In addition, in this embodiment, the flexible circuit board **24** and the side magnet **34** may have an overlapping region in the z-axis direction, to ensure that the side magnet **34** has a large size, and also increase a length of the flexible circuit board **24**, so that magnetic induction intensity of the magnetic circuit assembly **30** and vibration performance of the vibration assembly are improved. During specific implementation, in a negative direction of the y-axis, the straight line segment of the second connection part **242** of the flexible circuit board **24***a* exceeds the straight line segment of the first connection part **241**. In other words, relative to the head end of the first connection part **241**, the tail end of the second connection part **242** is disposed closer to the first corner part A**1** of the voice coil **23**. On the outer side of the first side surface **331** of the central magnet **33**, a first avoidance groove **343** is disposed on a surface that is of the first branch **341** and that faces the side electrode plate **32**. The tail end of the second connection part **242** of the flexible circuit board **24***a* may extend above the first branch **341**, and is disposed opposite to the first avoidance groove **343**. In addition, a first avoidance hole **323** is disposed at a location that is of the side electrode plate **32** and that corresponds to the first avoidance groove **343**, and the first avoidance hole **323** is connected to the first notch **321**. In this way, when the voice coil **23** drives the flexible circuit board **24***a* to vibrate in the z direction, the first avoidance groove **343** and the first avoidance hole **323** may provide avoidance space for movement of the second connection part **242**, to avoid interference caused by the first branch **341** and the side electrode plate **32** to movement of the second connection part **242**, and improve vibration performance of the vibration assembly **30**.
(50) Similarly, in a positive direction of the y-axis, the straight line segment of the second connection part **242** of the flexible circuit board **24***b* exceeds the straight line segment of the first connection part **241**. In other words, relative to the head end of the first connection part **241**, the tail end of the second connection part **242** is disposed closer to the second corner part B**1** of the voice coil. On the outer side of the third side surface **333** of the central magnet **33**, a second avoidance groove **344** is disposed on a surface that is of the first branch **341** and that faces the side electrode plate **32**. The tail end of the second connection part **242** of the flexible circuit board **24***b* may extend above the first branch **341**, and is disposed opposite to the second avoidance groove **344**. A second avoidance hole **324** is disposed at a location that is of the side electrode plate **32** and that corresponds to the second avoidance groove **344**, and the second avoidance hole **324** is connected to the second notch **322**. In this way, when the voice coil **23** drives the flexible circuit board **24***b* to vibrate in the z direction, the second avoidance groove **344** and the second avoidance hole **324** may provide avoidance space for movement of the second connection part **242**, to avoid interference caused by the first branch **341** and the side electrode plate **32** to movement of the second connection part **242**, and improve vibration performance of the vibration assembly **30**.
(51) It should be noted that, to reduce difficulty of a manufacturing process of the side magnet **34**, in some implementations, a part that is of the first branch **341** and that is provided with the first avoidance groove **343** (a third avoidance groove **344**) and a remaining part of the side magnet **34** may be separately formed, and then fixedly connected by bonding or the like.
(52) FIG. **11** is a sectional view of the electroacoustic transducer shown in FIG. **4** in a B-B direction. For example, an end that is of the central magnet **33** and that is close to the central electrode plate **31** is an N pole, and an end that is of the central magnet **33** and that is close to the lower electrode plate **35** is an S pole. An end that is of the side magnet **34** and that is close to the side electrode plate **32** is an S pole, and an end that is of the side magnet **34** and that is close to the lower electrode plate **35** is an N pole. A path of a magnetic line (shown by a dashed line in FIG. **11**) is “the N pole of the central magnet **33**—the central electrode plate **31**—the second gap **38**—the side electrode plate **32**—the S pole of the side magnet **34**—the N pole of the side magnet **34**—the lower electrode plate **35**—the S pole of the central magnet **33**”. The voice coil **23** is partially located in the second gap **38**, and the magnetic circuit assembly forms a magnetic field in the second gap **36**. Therefore, when the voice coil **23** is powered on, an ampere force in the z-axis direction may be generated, to drive the voice coil **23** and the sound diaphragm **21** connected to the

voice coil **23** to vibrate. During vibration, the sound diaphragm **21** pushes air in front (that is, in the positive direction of the z axis) of the sound diaphragm **21** to generate a sound wave, so that the electroacoustic transducer **510** emits a sound.

(53) In addition, when the voice coil **23** vibrates, the first connection part **241** and the second connection part **242** of the flexible circuit board **24***a* vibrate in the z-axis direction in the first notch **321** and the corresponding connection space **37**, and amplitudes gradually decrease from the head end of the first connection part **241** to the tail end of the second connection part **242**. The third connection part **243** of the flexible circuit board **24***a* is fastened to an edge of the side electrode plate **32**, and does not vibrate with the voice coil **23**. The auxiliary diaphragm **25** vibrates in the z-axis direction in the connection space **37** below the first notch **321**. The first connection part **241** and the second connection part **242** of the flexible circuit board **24***b* vibrate in the z-axis direction in the second notch **322** and the corresponding connection space **37**, and amplitudes gradually decrease from the head end of the first connection part **241** to the tail end of the second connection part **242**. The third connection part **243** of the flexible circuit board **24***b* is fastened to an edge of the side electrode plate **32**, and does not vibrate with the voice coil **23**. The auxiliary diaphragm **25** vibrates in the z-axis direction in the connection space **37** below the second notch **322**.

(54) In this embodiment, the sound diaphragm **21** located above the voice coil **23** may form a first compliant system of the electroacoustic transducer **510**, and the flexible circuit board **24** and the auxiliary diaphragm **25** located below the voice coil **23** may form a second compliant system of the electroacoustic transducer **510**. When the two compliant systems vibrate with the voice coil **23**, rolling vibration of the voice coil **23** can be suppressed, and the electroacoustic transducer **510** can obtain better sound quality. A compliance coefficient of the compliant system is a reciprocal of an elasticity coefficient, and a component with a higher compliance coefficient is more likely to deform under a force.

(55) For example, in the two compliant systems, the sound diaphragm **21** has high hardness and low compliance, so that the sound diaphragm **21** can smoothly push air to emit a sound. The flexible circuit board **24** and the auxiliary diaphragm **25** have low hardness and high compliance, so that total hardness of the two compliant systems is appropriate, to ensure working reliability of the electroacoustic transducer **510**. In addition, during specific design, hardness of the auxiliary diaphragm **25** may be greater than hardness of the flexible circuit board **24**, so that the second compliant system has higher stability, to better suppress swinging of the voice coil **23**, and reduce a rolling amplitude of the voice coil **23**.

(56) Compared with a conventional electroacoustic transducer in which a side magnet is arranged on a side surface of a central magnet, in the electroacoustic transducer in this embodiment, the side magnet is designed as an L-shaped structure, so that the two side magnets are disposed around the two diagonally arranged corner parts of the central magnet, to fill a magnet loss on outer peripheral sides of the two corner parts, and increase an overall size of the side magnets by at least 20%. Therefore, magnetic induction intensity of the magnetic circuit assembly can be effectively improved, and a driving capability of the magnetic circuit assembly is improved, so that the electroacoustic transducer has better sensitivity. It is verified that sensitivity of the electroacoustic transducer provided in this embodiment can be improved by more than 1 dB.

(57) The foregoing descriptions are merely specific implementations of this application, but are not intended to limit the protection scope of this application. Any variation or replacement readily figured out by a person skilled in the art within the technical scope disclosed in this application shall fall within the protection scope of this application. Therefore, the protection scope of this application shall be subject to the protection scope of the claims.

Claims

1. An electroacoustic transducer, comprising a central magnet, two side magnets, a lower electrode plate, a side electrode plate, a frame, a sound diaphragm, a voice coil, and two flexible circuit boards, wherein the central magnet and the two side magnets are separately disposed on a surface of the lower electrode plate, the central magnet is of a rectangular structure, and comprises a first corner part and a second corner part that are diagonally disposed, and a third corner part and a fourth corner part that are diagonally disposed, the two side magnets are disposed centrosymmetrically around the first corner part and the second corner part respectively, and a first gap exists between the two side magnets and the central magnet; the side electrode plate is disposed on surfaces of the two side magnets that are opposite to surfaces of the two side magnets that face the lower electrode plate; one surface of the frame is fixedly connected to a side that is of the side electrode plate and that is opposite to a side of the side electrode plate that faces the lower electrode plate, and another surface of the frame is fixedly connected to a periphery of the sound diaphragm; the voice coil is located inside the frame, the voice coil is of a rectangular ring structure, one end of the voice coil is fixedly connected to the sound diaphragm, and another end of the voice coil is inserted into the first gap; and the two flexible circuit boards are symmetrically distributed on peripheral sides of the voice coil, and each flexible circuit board is fixedly connected to the side electrode plate; the voice coil comprises a third corner and a fourth corner that correspond to the third corner part and the fourth corner part respectively; and the two flexible circuit boards are respectively disposed on outer peripheral sides of the third corner and the fourth corner, one end of each flexible circuit board is fixedly connected to the voice coil, and another end of each flexible circuit board is fixedly connected to a surface of the side electrode plate that faces the lower electrode plate.
2. The electroacoustic transducer according to claim 1, wherein the central magnet comprises a first side surface and a second side surface that are adjacent to the first corner part, and a third side surface and a fourth side surface that are adjacent to the second corner part; and each side magnet of the two side magnets comprises a first branch and a second branch that are connected to form an L-shaped structure, the first branch and the second branch of a first side magnet of the two side magnets are respectively located on outer sides of the first side surface and the second side surface, and the first branch and the second branch of a second side magnet of the two side magnets are respectively located on outer sides of the third side surface and the fourth side surface.
3. The electroacoustic transducer according to claim 2, wherein and the first branch of the first side magnet extends between the first corner part and the third corner part, and the second branch of the first side magnet extends to an end that is of the second side surface and that is adjacent to the fourth corner part; and the first branch of the second side magnet extends between the second corner part and the fourth corner part, and the second branch of the second side magnet extends to an end that is of the fourth side surface and that is adjacent to the third corner part.
4. The electroacoustic transducer according to claim 3, wherein each of the two flexible circuit boards comprises a first connection part, a second connection part, and a third connection part that are sequentially connected, each first connection part is fixedly connected to the voice coil, each second connection part is located on a side that is of the corresponding first connection part and that is opposite to a side of the corresponding first connection part that faces the voice coil, each third connection part is located on a side that is of the second connection part and that is opposite to a side of the corresponding second connection part that faces the first connection part, and each third connection part is fixedly connected to the side electrode plate; and the side electrode plate is provided with a first notch and a second notch that are symmetrically distributed, and the first notch and the second notch respectively expose first connection parts and a part of second connection parts of the two flexible circuit boards.
5. The electroacoustic transducer according to claim 4, wherein for each flexible circuit board, an end that is of the corresponding second connection part and that is connected to the corresponding

third connection part is stacked on a surface that is of a closest first branch and that is opposite to a surface of the closest first branch that faces the lower electrode plate, and the second connection parts of the two flexible circuit boards respectively extend on an outer peripheral side of the first side surface and an outer peripheral side of the third side surface, and an avoidance groove is disposed at a location that is on the closest first branch and that corresponds to an end part of the corresponding second connection part.

6. The electroacoustic transducer according to claim 5, wherein avoidance holes are respectively disposed at locations that are of the side electrode plate and that correspond to two avoidance grooves, and the avoidance holes are connected to the first notch and the second notch respectively.

7. The electroacoustic transducer according to claim 4, wherein the electroacoustic transducer further comprises two auxiliary diaphragms, the two auxiliary diaphragms are respectively disposed on sides that are of the two flexible circuit boards and that face the sound diaphragm, one end of each auxiliary diaphragm is fixedly connected to the first connection part of a corresponding flexible circuit board, and the other end of the each auxiliary diaphragm is fixedly connected to the third connection part of the corresponding flexible circuit board.

8. The electroacoustic transducer according to claim 7, wherein the sound diaphragm comprises a first vibration part, a cross-sectional shape of the first vibration part of the sound diaphragm is an arc shape, and the first vibration part of the sound diaphragm protrudes in a direction opposite to the lower electrode plate; and each auxiliary diaphragm comprises a second vibration part, a cross-sectional shape of each second vibration part of the each auxiliary diaphragm is an arc shape, and the second vibration part of the each auxiliary diaphragm protrudes in a direction facing the lower electrode plate.

9. The electroacoustic transducer according to claim 7, wherein a third notch and a fourth notch are respectively disposed at locations that are of the lower electrode plate and that correspond to the two auxiliary diaphragms.

10. The electroacoustic transducer according to claim 1, wherein the electroacoustic transducer further comprises a connection frame, the connection frame is located between the voice coil and the sound diaphragm, one end of the connection frame is fixedly connected to the voice coil, and another end of the connection frame is fixedly connected to the sound diaphragm.

11. An electronic device, comprising a housing, a display module, and a receiver, wherein the receiver is disposed in the housing, wherein the display module comprises a glass cover and a display panel, the glass cover is fastened to the housing, and the display panel is fastened to an inner surface that is of the glass cover and that faces the housing; a receiver hole is disposed on the glass cover, or a receiver hole is formed between an edge of the glass cover and the housing, or a receiver hole is disposed on the housing; and the receiver comprises a central magnet, two side magnets, a lower electrode plate, a side electrode plate, a frame, a sound diaphragm, a voice coil, and two flexible circuit boards, wherein the central magnet and the two side magnets are separately disposed on a surface of the lower electrode plate, the central magnet is of a rectangular structure, and comprises a first corner part and a second corner part that are diagonally disposed, and a third corner part and a fourth corner part that are diagonally disposed, the two side magnets are disposed centrosymmetrically around the first corner part and the second corner part respectively, and a first gap exists between the two side magnets and the central magnet; the side electrode plate is disposed on surfaces that are of the two side magnets and that are opposite to surfaces of the two side magnets that face the lower electrode plate; one surface of the frame is fixedly connected to a side that is of the side electrode plate and that is opposite to a side of the side electrode plate that faces the lower electrode plate, and another surface of the frame is fixedly connected to a periphery of the sound diaphragm; the voice coil is located inside the frame, the voice coil is of a rectangular ring structure, one end of the voice coil is fixedly connected to the sound diaphragm, and another end of the voice coil is inserted into the first gap; and the two flexible circuit boards are symmetrically distributed on a peripheral side of the voice coil, and each flexible circuit board is

fixedly connected to the side electrode plate, and a sound emitted by the receiver is transmitted to an outer side of the electronic device through the receiver hole; the voice coil comprises a third corner and a fourth corner that correspond to the third corner part and the fourth corner part respectively; and the two flexible circuit boards are respectively disposed on outer peripheral sides of the third corner and the fourth corner, one end of each flexible circuit board is fixedly connected to the voice coil, and another end of each flexible circuit board is fixedly connected to a surface of the side electrode plate that faces the lower electrode plate.
12. The electronic device according to claim 11, wherein the central magnet comprises a first side surface and a second side surface that are adjacent to the first corner part, and a third side surface and a fourth side surface that are adjacent to the second corner part; and each side magnet of the two side magnets comprises a first branch and a second branch that are connected to form an L-shaped structure, the first branch and the second branch of a first side magnet of the two side magnets are respectively located on outer sides of the first side surface and the second side surface, and the first branch and the second branch of a second side magnet of the two side magnets are respectively located on outer sides of the third side surface and the fourth side surface.
13. The electronic device according to claim 12, wherein the first branch of the first side magnet extends between the first corner part and the third corner part, and the second branch of the first side magnet extends to an end that is of the second side surface and that is adjacent to the fourth corner part; and the first branch of the second side magnet extends between the second corner part and the fourth corner part, and the second branch of the second side magnet extends to an end that is of the fourth side surface and that is adjacent to the third corner part.
14. The electronic device according to claim 13, wherein each of the two flexible circuit boards comprises a first connection part, a second connection part, and a third connection part that are sequentially connected, each first connection part is fixedly connected to the voice coil, each second connection part is located on a side that is of the corresponding first connection part and that is opposite to a side of the corresponding first connection part that faces the voice coil, each third connection part is located on a side that is of the corresponding second connection part and that is opposite to a side of the corresponding second connection part that faces the first connection part, and each third connection part is fixedly connected to the side electrode plate; and the side electrode plate is provided with a first notch and a second notch that are symmetrically distributed, and the first notch and the second notch respectively expose first connection parts and a part of second connection parts of the two flexible circuit boards.
15. The electronic device according to claim 14, wherein for each flexible circuit board, an end that is of the corresponding second connection part and that is connected to the corresponding third connection part is stacked on a surface that is of closest first branch and that is opposite to a surface of the closest first branch that faces the lower electrode plate, and the second connection parts of the two flexible circuit boards respectively extend on an outer peripheral side of the first side surface and an outer peripheral side of the third side surface, and an avoidance groove is disposed at a location that is on the closest first branch and that corresponds to an end part of the corresponding second connection part.
16. The electronic device according to claim 15, wherein avoidance holes are respectively disposed at locations that are of the side electrode plate and that correspond to two avoidance grooves, and two avoidance holes are connected to the first notch and the second notch respectively.
17. The electronic device according to claim 14, wherein the receiver further comprises two auxiliary diaphragms, the two auxiliary diaphragms are respectively disposed on sides that are of the two flexible circuit boards and that face the sound diaphragm, one end of each auxiliary diaphragm is fixedly connected to the first connection part of a corresponding flexible circuit board, and another end of the each auxiliary diaphragm is fixedly connected to the third connection part of the corresponding flexible circuit board.
18. The electronic device according to claim 17, wherein the sound diaphragm comprises a first

vibration part, a cross-sectional shape of the first vibration part of the sound diaphragm is an arc shape, and the first vibration part of the sound diaphragm protrudes in a direction opposite to the lower electrode plate; and each auxiliary diaphragm comprises a second vibration part, a cross-sectional shape of each second vibration part of the each auxiliary diaphragm is an arc shape, and the second vibration part of the each auxiliary diaphragm protrudes in a direction facing the lower electrode plate.

19. The electroacoustic transducer according to claim 1, further comprising: a central electrode plate over the central magnet, wherein the side electrode plate encircles the central electrode plate.

20. The electronic device according to claim 11, wherein the receiver further comprises: a central electrode plate over the central magnet, wherein the side electrode plate encircles the central electrode plate.